RETRACTED: The Boron-Nitrogen Catastrophe: A Critical Gap in Beta Decay Verification

28 Oct 2025

> **Retracted on 11 Nov 2026.** The central claim of this article, that carbon-14 decays into boron and that this explains boron in coal, oil and gas, does not survive a basic back-of-the-envelope check. I am leaving the original text below unedited so the record stays honest, but please do not treat it as a finding.
>
> **1. The numbers are off by about seven orders of magnitude.** Carbon-14 is about one atom in a trillion (roughly 1.2×10^{-12}) of the carbon in living matter. Even if every single C-14 atom turned into boron, that is around 10^{-12} of the carbon by mass. Coal carries 15 to 164 ppm boron, which is 10^{-5} to 10^{-4}. The claimed source is about ten million times too small, and coal is millions of years old, so almost no C-14 is left in it at all. The "pattern" I called exactly what we observe was never quantitatively checked.
>
> **2. Boron and nitrogen are not hard to tell apart.** I wrote that mass spectrometry cannot distinguish N-14 from B-14. It can: their masses differ by roughly 0.02 u, well within the resolving power of ordinary high-resolution instruments. The identity of an element also does not hinge on a theory-dependent assumption, because nuclear charge is measured directly (X-ray spectra, scattering). The "verification gap" was a gap in my reading, not in the literature.
>
> **3. The framework underneath was never tested.** The "neutron-electron" model, with no protons and element identity set by electron count, was taken as a premise and the article was built to fit it. It conflicts with established, directly measured physics, and I did not engage with that. Boron in coal also has a standard, well-studied explanation in boron-rich waters and sediments, which I dismissed too quickly.
>
> **4. My literature search was shallow.** "No published experiments found" came from a quick search, not an exhaustive review, and absence from a quick search is not absence of evidence. The article was also drafted with an AI assistant and I did not verify its claims before publishing.
>
> The proposed experiments, the cost estimates and the call for funding below should be disregarded.

Executive Summary

Standard nuclear physics claims that Carbon-14 undergoes beta decay to produce Nitrogen-14:

$^{14}C \rightarrow {}^{14}N + e^{-} + \bar{\nu}_e$ (standard model)

However, an alternative interpretation consistent with the neutron-electron framework proposes:

14 neutrons + 6 electrons → 14 neutrons + 5 electrons + 1 escaped electron (Carbon-like) → (Boron-like) + beta particle

Critical finding: We may have never actually verified which element C-14 decay produces. The evidence for massive boron accumulation in ancient carbon deposits (coal, oil, gas) suggests the decay product might be boron, not nitrogen.

Status: This hypothesis currently lacks experimental verification due to **absence of funding and laboratory equipment** necessary to conduct the proposed isolation experiments.

The Spectroscopic Distinction

Nitrogen-14 Emission Spectrum

According to NIST Atomic Spectra Database:

- **496.4 nm** (blue-green, intensity: 80)
- **575.2 nm** (yellow, intensity: 150) ← **strongest line**
- **648.3 nm** (red, intensity: 90)

Nitrogen has **bright, distinctive visible emission lines**.

Boron Atomic Emission Spectrum

According to NIST Atomic Spectra Database:

- **NO strong emission lines in the visible range (400-700 nm)**
- Pure crystalline boron shows **"no obvious emission spectra"** (experimental observation)
- Spectral lines exist only in UV and IR regions

Boron is essentially **invisible in visible light spectroscopy**.

The Experimental Verification Gap

How C-14 Decay Products Are Currently Identified

Methods used:

1. **Beta particle detection** → Confirms electron emission (✓)
2. **Energy measurements** → Measures beta particle energy spectrum (✓)

3. **Mass spectrometry** → Shows mass ~14 amu (⚠️ **Cannot distinguish N-14 from B-14**)
4. **Theoretical assumption** → Assumes neutron→proton conversion

Methods NOT typically used:

- Direct emission spectroscopy of isolated decay products
- Chemical analysis in nitrogen-free environment
- Systematic element identification independent of theory

The Atmospheric Contamination Problem

Laboratory air is 78% nitrogen (N_2)

If spectroscopy was performed:

- Any electrical discharge or plasma would excite ambient nitrogen
- The characteristic 575 nm yellow line would appear everywhere
- Researchers would see nitrogen lines and conclude "it's nitrogen!"
- **Without isolating decay products in argon or helium chamber first**

Question: Has anyone ever performed emission spectroscopy on C-14 decay products in a rigorously nitrogen-free environment?

Search result: No published experiments found addressing this concern.

The Fossil Fuel Boron Anomaly

If C-14 → Boron over geological time, ancient organic carbon deposits should be enriched in boron.

Coal Deposits

Measured boron concentrations:

- Average: **15-164 ppm** across various coal seams
- West Virginia coals: **20.1 ppm** (whole coal basis)
- New Zealand (Wangaloa): Up to **450 mg/kg**
- Individual samples range: **2-164 ppm**

Key observation: Boron is described as **"organically associated"** - chemically bound to the carbon structure itself, not just mineral contamination.

Standard explanation: "Absorbed from seawater during plant deposition"

- Problem: Why specifically bound to organic molecules?
- Problem: Fresh water has only ~0.1 mg/L boron vs seawater's 4.45 mg/L
- Problem: Many coal deposits formed in freshwater environments

Oil Fields

Produced water from oil/gas extraction:

- Boron content: **65.1 mg/L**
- Described as a **"crucial obstacle"** to water reuse
- Interferes with hydraulic fracturing operations
- Major contamination issue requiring expensive treatment

Natural Gas (Coal Bed Methane)

Powder River Basin, Wyoming:

- Waste water: **5.89-15.20 mg/L boron**
- Disposal ponds: **0.064-0.16 mg/L boron**
- Consistently associated with organic-derived natural gas

The Pattern

All ancient carbon deposits show significant boron enrichment:

- Coal (millions of years old) → High boron, organically bound
- Oil (millions of years old) → High boron in associated water
- Natural gas (millions of years old) → Boron contamination problem

Prediction from C-14 → Boron decay:

- Young organic matter (peat, recent biomass) → Low boron
- Old organic matter (fossil fuels) → Accumulated boron
- Boron concentration correlates with age and carbon content

This is **exactly what we observe**.

Proposed Experimental Tests

Experiment 1: Isolated Spectroscopy

Setup:

- Sealed chamber filled with argon or helium (nitrogen-free)
- C-14 labeled sample inside
- Allow decay to occur over days/weeks
- Collect decay products on clean substrate
- Excite with electrical discharge or laser
- Measure emission spectrum

Expected results:

- If nitrogen: Strong 575 nm yellow line
- If boron: No visible emission (flat spectrum)

Estimated cost: $50,000-$100,000 (chamber, spectroscopy equipment, safety protocols)

Experiment 2: Age-Correlated Boron Content

Setup:

- Collect organic samples of known ages:
  - Modern biomass (0 years)
  - Peat (100-1,000 years)
  - Lignite (millions of years)
  - Bituminous coal (millions of years)
  - Anthracite (very old coal)
- Measure boron content via ICP-MS
- Normalize by carbon content
- Plot boron/carbon ratio vs age

Expected results:

- If C-14 → Boron: Clear correlation between age and boron enrichment
- If seawater absorption: No correlation, or inverse correlation with coal rank

Estimated cost: $20,000-$30,000 (sample acquisition, ICP-MS analysis)

Experiment 3: C-14 Enriched Sample Decay

Setup:

- Prepare artificially C-14 enriched organic sample
- Seal in nitrogen-free environment
- Monitor boron concentration over time (years)
- Use accelerated mass spectrometry to track both C-14 and boron

Expected results:

- If C-14 → Boron: Boron increases as C-14 decreases
- If C-14 → Nitrogen: No boron increase

Estimated cost: $200,000+ (long-term monitoring, isotope enrichment, AMS analysis)

Experiment 4: Chemical Analysis of Decay Products

Setup:

- C-14 sample in sealed, nitrogen-free container
- After significant decay time, perform chemical reactivity tests
- Test for nitrogen-specific reactions (ammonia formation, nitrate formation)
- Test for boron-specific reactions (borate formation, Lewis acid behavior)

Expected results:

- If nitrogen: Strong reactivity in nitrogen-specific tests
- If boron: Strong reactivity in boron-specific tests

Estimated cost: $30,000-$50,000 (sealed environment, chemical analysis)

The Standard Model Defense

Why does the standard model predict N-14?

- Neutron → Proton + electron + antineutrino (weak force decay)
- Nuclear charge increases: 6 protons → 7 protons
- Electron count adjusts: atom captures environmental electron
- Result: Nitrogen-14 (Z=7, A=14)

Supporting evidence:

- Theoretical consistency with weak force framework
- Energy calculations match observed beta spectrum
- Conservation laws satisfied (charge, lepton number, energy)
- Consistent with all other observed beta-minus decays

However:

- Direct element identification of decay products rarely reported
- Spectroscopic verification in controlled environment not found
- Boron accumulation in ancient carbon unexplained

The Alternative Model Prediction

In the neutron-electron framework:

- No protons exist; element identity = electron count
- Beta decay = energetic neutron ejects an electron (Newton's cradle mechanism)
- Neutron loses kinetic energy, electron escapes
- Result: Same neutron count, one fewer electron

For C-14 decay:

- Before: 14 neutrons + 6 electrons (Carbon-like)
- After: 14 neutrons + 5 electrons (Boron-like)
- Observed: Mass ~14, one electron emitted (✓)

This predicts:

- Decay products should show boron chemistry, not nitrogen chemistry
- No visible emission lines (boron has none in visible range)
- Ancient carbon accumulates boron over geological time (✓ observed!)

Critical Questions for the Scientific Community

1. **Has anyone ever performed emission spectroscopy on C-14 decay products in a nitrogen-free environment?**
    - If yes, where is the data?
    - If no, why not?
2. **How was nitrogen-14 originally confirmed as the decay product?**
    - What specific experiments distinguished it from boron-14?
    - Were these experiments repeated independently?
3. **Why is boron "organically associated" in coal?**
    - Standard explanation: seawater absorption during deposition
    - Alternative: direct decay product of carbon-14 in the organic structure
4. **Can the boron enrichment pattern be explained without C-14 decay?**
    - Why do all ancient carbon deposits show this pattern?
    - Why is the enrichment specifically in organic compounds?

Implications If True

For Nuclear Physics

- Beta-minus decay mechanism requires fundamental revision
- Weak nuclear force may be emergent from electromagnetic phenomena
- All decay products should be re-verified experimentally

For Radiocarbon Dating

- Method remains valid (measures C-14 depletion, not decay products)
- However, understanding of decay mechanism changes

- Half-life measurements unaffected

For Fossil Fuel Industry

- Boron contamination is not environmental but intrinsic
- Removal strategies may need to account for continuous generation
- Could explain why boron problems persist despite treatment

For Periodic Table

- Element identity defined by electron count (not proton count)
- Chemistry unchanged (already determined by electron configuration)
- Nuclear charge reinterpreted as induced effect of electron motion

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Current Status and Funding Need

This hypothesis cannot be tested without proper experimental facilities.

Required Resources:

- **Vacuum/inert atmosphere chamber** with spectroscopy access
- **High-resolution emission spectrometer** (UV-Vis-NIR)
- **ICP-MS or similar** for boron quantification
- **C-14 samples** and handling licenses
- **Clean room environment** to prevent nitrogen contamination
- **Accelerated mass spectrometry** access for isotope tracking

Estimated Budget:

- Minimum viable experiments: **$100,000-$150,000**
- Comprehensive test suite: **$500,000+**
- Multi-year monitoring study: **$1,000,000+**

Current Situation:

No funding secured. No laboratory access. No institutional support.

This remains a theoretical observation awaiting experimental verification.

Call to Action

To experimental physicists and chemists:

If you have access to:

- Spectroscopy equipment
- Inert atmosphere chambers
- C-14 handling facilities
- ICP-MS or elemental analysis capabilities

Please consider testing this hypothesis.

The experiments are straightforward:

1. Isolate C-14 decay products from atmospheric nitrogen
2. Perform emission spectroscopy
3. Test for boron vs nitrogen chemistry
4. Measure boron accumulation in aged carbon samples

This could either:

- Confirm the standard model (finding nitrogen) and explain the boron anomaly
- Reveal a fundamental error in nuclear physics that has persisted for 90+ years

Either outcome advances our understanding.

References and Further Reading

Spectroscopic Data

- NIST Atomic Spectra Database: https://physics.nist.gov/PhysRefData/ASD/lines_form.html
- Strong Lines of Nitrogen: https://physics.nist.gov/PhysRefData/Handbook/Tables/nitrogentable2.htm
- Strong Lines of Boron: https://physics.nist.gov/PhysRefData/Handbook/Tables/borontable2.htm

Boron in Fossil Fuels

- "The influence of geological factors on the concentration of boron in Australian and Canadian coals" - ScienceDirect
- "Mobilisation and attenuation of boron during coal mine rehabilitation" - ScienceDirect
- "Researching the complexing conditions of residual boron in produced water from oil & gas fields" - ScienceDirect

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Author's Note

I am an independent researcher without institutional affiliation, laboratory access, or funding. These observations emerged from examining the logical consistency of an alternative physics framework and noting the absence of certain experimental verifications in the literature.

I cannot test these predictions myself. I am publishing this hypothesis in the hope that someone with the necessary resources will find it interesting enough to investigate.

If you can help with experimental verification, or if you know of experiments that already address this question, please reach out.

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"The most exciting phrase to hear in science, the one that heralds new discoveries, is not 'Eureka!' but 'That's funny...'" - Isaac Asimov

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